

# Roger Richard Demello

AI Engineer | India | B.Tech 2026 | Immediate joiner | Any timezone or shift

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## PROFESSIONAL SUMMARY

AI engineer shipping agentic systems in production – multi-agent orchestration on LangGraph, Retrieval-Augmented Generation (RAG), multilingual voice agents, and a live conversational agent on WhatsApp and Instagram.

## TECHNICAL SKILLS

**Programming Languages:** Python, TypeScript, JavaScript, SQL, Bash, YAML, Java, C++, C  
**Agentic AI:** AI agents, agentic workflows, multi-agent systems, agent orchestration, LangGraph, LangChain, MCP  
**Generative AI and LLMs:** LLMs, RAG, prompt engineering, embeddings, semantic search, FAISS, ChromaDB, LlamaIndex  
**Voice and Multimodal:** speech-to-text, text-to-speech, Sarvam AI, Murf, Web Audio API, OCR, computer vision  
**Model Providers and Backend:** OpenAI, Anthropic Claude, Google Gemini, Ollama, FastAPI, Node.js, REST APIs, WebSockets  
**Production and Delivery:** Docker, CI/CD, GitHub Actions, AWS, telemetry, Sentry, pytest, Playwright, PostgreSQL

## WORK EXPERIENCE

**Innovun Global** August 2026 – Present

*AI Engineer Intern*

- Build a live multi-channel conversational agent on **LangChain** and **LangGraph** – **Cruz Roja Mexicana** (Red Cross, Mexico).
- Integrate the **WhatsApp Business Cloud API** and **Instagram Graph API** across **3 channels** into one agent schema.
- Develop a **natural-language-to-SQL agent** consolidating **2 enterprise source systems** into one queryable database.
- Troubleshoot production defects end to end and author **technical documentation** for client **stakeholders**.

**AI LifeBOT (B2B Startup)**

January 2026 – June 2026

*AI Engineer Intern*

- Architected **RAG pipelines** inside **3 production applications** serving **200+ users**, with real-time streaming.
- Sustained **99.5% uptime** with **telemetry** and audit logging, owning **incident response** and root cause analysis.
- Reduced query and response **latency 35%** by profiling an inherited codebase across the full request path.
- Conducted **code reviews** and automated **pytest** suites in **CI/CD**, shipping **5+ features** in an **Agile** team.

**CFM, RCOEM**

May 2025 – July 2025

*Machine Learning Research Intern*

- Engineered measurable features in **Python** and **Pandas** from **1,000+** raw, inconsistent health records.
- Trained and validated a **classification model**, reaching **87% accuracy** on held-out records.
- Raised result reliability **25%** through **cross-validation** across multiple splits, not one sample.
- Standardized **data pipelines** into **tested, reusable components**, cutting development time **30%**.

## PROJECTS

**DealSentry** | *TypeScript, React, Express.js, Prisma ORM, PostgreSQL, OpenAI API, Puppeteer (PDF)* | [Live](#) | [GitHub](#)

- Shipped a live **AI-assisted** review platform on the **OpenAI API**, auto-parsing uploaded documents on upload.
- Routed approvals through structured reviews with **SLA tracking** and **audit logging** over **PostgreSQL**.
- Integrated **Salesforce, HubSpot, Gmail** and **Google Drive** APIs through documented data interfaces.
- Enforced **role-based access control** across every review, scoping each action to the assigned user role.

**contentflow-ai** | *Python, LangGraph, FastAPI, Streamlit, Azure OpenAI, Redis, Docker, pytest* | [GitHub](#)

- Architected a **multi-agent LangGraph pipeline** automating drafting, compliance review, localization and publishing.
- Gated every publish behind a **human-in-the-loop approval** step before anything reached a live channel.
- **Validated** each draft with policy-aware **RAG** over policy documents, auto-remediating and logging violations.

**Autonomous Executive Email Copilot** | *Python, FastAPI, Pydantic, React, PostgreSQL, OpenAI, Docker* | [Live](#) | [GitHub](#)

- Benchmarked **4 agent policy modes** – heuristic, perturbation, LLM-driven and multi-agent hybrid – on one harness.
- Instrumented **telemetry, episode replay** and alert evaluation around a long-running **agent loop**.
- Designed a deterministic **evaluation harness** scoring agents on bounded, numerically stable grading metrics.
- Published benchmark results with honest findings, including the runs where a policy underperformed its baseline.

## EDUCATION

**Shri Ramdeobaba College of Engineering and Management (RCOEM)**

2022 – 2026

B.Tech, Electronics & Communication Engineering

**GPA: 8.90/10**

Minor: **Artificial Intelligence & Machine Learning**

**GPA: 9.67/10**

## CERTIFICATIONS AND AWARDS

**AWS Certified Cloud Practitioner** – Amazon Web Services (October 2025)

**Hackathon Finalist** – Paytm x Sarvam x Logitech AI National Hackathon

**Second Place** – ByteSize Sage AI National Hackathon